# Assignment 2

## Goal:

The goal of this assignment is to develop predictive models for patient mortality and length of stay that adhere to hospital alarm protocols, alongside clustering techniques to uncover underlying clinical patient profiles. It also aims to use explainable AI to interpret these models and provide an evidence-based recommendation regarding their ethical suitability for real-world hospital deployment.

## Instructions for Students:

- You may use Python or Orange Canvas for this assignment.
- If using Python, you must set the random_seed based on your wnumber for all data splits and model initializations.
  For example, if wnumber is w1723axa use random_seed = 1723.
- You **may** use Generative AI to assist with writing code or navigating the Orange interface. However, your report must be your own analysis. To receive credit, all answers must directly reference the specific numerical outputs, plots, and implications generated by your unique seed.
- Your report **must follow the exact numbering of questions provided in this assignment**. Failure to follow the specified numbering will result in **no points being awarded for the corresponding responses**.
- Include your **Full Name** and **WNumber** in the header of every page of your report.
- Submit your **report as a PDF file**.
- If you use **Python or MATLAB**, submit all relevant **code files** along with your report.
- If you use **Orange**, submit a **separate file containing screenshots** of your workflow and results. Do not include the screenshots file as part of the report.
- Clearly document **any assumptions, preprocessing decisions, or other decisions you made** during the analysis in your report.

# Part 1 – Supervised Learning

## Due: MM-DD-YYYY

### A. Classification

i. Train any 2 models. Both models must use the **same features and the same train/test split**, so that differences between them come from the models, not the data. Keep both trained models: you will explain and compare them in Part 3.

ii. Report accuracy, recall, precision, specificity and AUROC for both of your models at the default threshold of 0.5. Which metric would you use and why?

iii. Then we need to define the ratio:
**False alarms per true alert = False Positives ÷ True Positives (FP ÷ TP)**
Assume that the hospital protocol allows a maximum of 3 false alarms per true alert (3:1).

For each of your two models:

1. Report the confusion matrix at the default threshold of 0.5 and calculate the false alarm ratio from the exact FP and TP counts. Show your work.
2. State whether the model meets the 3:1 protocol.
3. Find the **lowest** threshold that still meets the protocol. Report the new threshold, confusion matrix, false alarm ratio, and change in recall compared with 0.5. In plain terms, how many more (or fewer) deaths are now flagged, and how many more (or fewer) alerts must the nurses check?
4. If no threshold meets the protocol without recall falling close to zero, report this and explain what it tells you about the model.

### B. Regression

i. A child who dies on day 2 has a short length of stay, as does a child who recovers quickly. Train your regression model twice: once on all children, and once excluding children who died. Report both sets of results and explain the difference using your own numbers. Report MAE, RMSE and $R^2$ for both runs. Which version would a hospital bed-manager actually want, and why?

ii. Analyze your residual plot and identify the specific range of the target variable where your model makes the largest magnitude of errors (under-predicting or over-predicting).

# Part 2 – Unsupervised Learning

## Due: MM-DD-YYYY

Drop all target variables before clustering: in-hospital mortality, length of stay. The targets must not appear in any clustering run or cluster profile.

A. List the features you used and explain how you scaled or encoded each type: continuous measurements, categorical variables, and yes/no test results. Explain why scaling matters for distance-based clustering, using one feature from this dataset as an example of what would go wrong without it.
B. Run K-Means with k = 2, 3 and 4. You can experiment with different features if you are not happy with the clusters. For your chosen feature set, describe what happens when you go from k = 2 to 3 to 4. Does an existing cluster split into two, or do the clusters reorganize completely? Explain what this tells you about the structure of the data.
C. For your chosen model, extract the centroid for each cluster: the average value of each feature. Present the centroids in the **original units**, not scaled values, and explain why the original units are needed for clinical interpretation.
D. Write a 1–2 sentence clinical profile for each cluster, based strictly on its defining features. For example: "Cluster 1 represents older children with elevated respiratory rates and low oxygen saturation." Each profile should name the 2–3 features that most distinguish that cluster from the others, and every claim must be supported by numbers from your centroid table.

# Part 3 – Results/ explainable AI

## Due: MM-DD-YYYY

A. Compute feature importance with two different methods for two classification models. Identify where the rankings disagree in your results and explain why those two methods would disagree on those particular features. Compare the important features across 2 models and two different methods.
B. Pick one child your model correctly flagged and one child who died but was missed, and report their row IDs. Produce a local explanation for each. For the missed child, identify which features pushed the prediction towards survival.
C. In no more than 300 words, write a recommendation to a hospital ethics board on whether your model should be piloted in a local hospital. Use only evidence from

Parts 1-3 of your own report. Generic statements about AI fairness or transparency, without reference to your own results, will receive no marks.

| Part | Marks |
|---|---|
| Part 1: Supervised Learning | 40 |
| Part 2: Unsupervised Learning | 28 |
| Part 3: Explainable AI | 27 |
| Report quality and compliance | 5 |
| **Total** | **100** |